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DSCI 552
HW 6

6.6.3

a. the training RSS will steadily decrease. since increasing s adds more predictors back in lasso regularization, all the way until the least squares solution exists in the constraint, any model with a larger s will be better able to fit the training data than a model with a smaller s until the LS solution, which produces an optimal training RSS.

b. the test RSS will display the classic decrease-then-increase curve as increasing s will reduce underfitting at first (decreasing test RSS) before the model begins overfitting (increasing test RSS) as more and more predictors are used in the model.

c. variance will increase as s increases, since a larger s allows for a larger class of models (more predictors) to be used as the fitted model.

d. (squared) bias will decrease as s increases as increasing the number of predictors decreases the chance that the model will underfit due to the optimal model not being in the class of models allowed as a fitted model. this is the bias-variance tradeoff.

e. the irreducible error ε will remain constant, as it is not a function of s.

6.6.5

a. the ridge regression optimization problem is to minimize

$$\sum_{i=1}^{2}\left(y_i - \beta_0 - \sum_{j=1}^{2}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{2}\beta_j^2$$

then we are minimizing the equation

$$(y_1 - \beta_0 - \beta_1 x_{11} - \beta_2 x_{12})^2 + (y_2 - \beta_0 - \beta_1 x_{21} - \beta_2 x_{22})^2 + \lambda(\beta_1^2 + \beta_2^2)$$

in this setting.

b. let $f = (y_1 - \hat{\beta}_0 - \hat{\beta}_1 x_{11} - \hat{\beta}_2 x_{12})^2 + (y_2 - \hat{\beta}_0 - \hat{\beta}_1 x_{21} - \hat{\beta}_2 x_{22})^2 + \lambda(\hat{\beta}_1^2 + \hat{\beta}_2^2)$

we know $\hat{\beta}_0 = 0$, $x_{11} = x_{12}$, $x_{21} = x_{22}$, $x_{11} = -x_{21} = -x_{22}$, so simplify:

$$f = (y_1 - \hat{\beta}_1 x_{11} - \hat{\beta}_2 x_{11})^2 + (y_2 + \hat{\beta}_1 x_{11} + \hat{\beta}_2 x_{11})^2 + \lambda(\hat{\beta}_1^2 + \hat{\beta}_2^2)$$

also, $y_2 = -y_1$, so:

$$f = (y_1 - \hat{\beta}_1 x_{11} - \hat{\beta}_2 x_{11})^2 + (-y_1 + \hat{\beta}_1 x_{11} + \hat{\beta}_2 x_{11})^2 + \lambda(\hat{\beta}_1^2 + \hat{\beta}_2^2)$$

$$= (y_1 - \hat{\beta}_1 x_{11} - \hat{\beta}_2 x_{11})^2 + ((-1)(y_1 - \hat{\beta}_1 x_{11} - \hat{\beta}_2 x_{11}))^2 + \lambda(\hat{\beta}_1^2 + \hat{\beta}_2^2)$$

$$= 2(y_1 - \hat{\beta}_1 x_{11} - \hat{\beta}_2 x_{11})^2 + \lambda(\hat{\beta}_1^2 + \hat{\beta}_2^2)$$

$$= 2(y_1^2 - \hat{\beta}_1 x_{11} y_1 - \hat{\beta}_2 x_{11} y_1 - y_1\hat{\beta}_1 x_{11} + \hat{\beta}_1^2 x_{11}^2 + \hat{\beta}_1\hat{\beta}_2 x_{11}^2 - \hat{\beta}_2 x_{11} y_1 + \hat{\beta}_1\hat{\beta}_2 x_{11}^2 + \hat{\beta}_2^2 x_{11}^2) + \lambda\hat{\beta}_1^2 + \lambda\hat{\beta}_2^2$$

$$= 2y_1^2 - 4\hat{\beta}_1 x_{11} y_1 - 4\hat{\beta}_2 x_{11} y_1 + 2\hat{\beta}_1^2 x_{11}^2 + 2\hat{\beta}_2^2 x_{11}^2 + 4\hat{\beta}_1\hat{\beta}_2 x_{11}^2 + \lambda\hat{\beta}_1^2 + \lambda\hat{\beta}_2^2$$

then we find $\begin{bmatrix}\hat{\beta}_1 \\ \hat{\beta}_2\end{bmatrix}$ which minimize f by taking partial derivatives and solving:

$$\frac{\partial f}{\partial \hat{\beta}_1} = -4x_{11}y_1 + 4x_{11}^2\hat{\beta}_1 + 4x_{11}^2\hat{\beta}_2 + 2\lambda\hat{\beta}_1 = 0$$

$$4x_{11}^2\hat{\beta}_1 + 2\lambda\hat{\beta}_1 = 4x_{11}y_1 - 4x_{11}^2\hat{\beta}_2$$

$$\hat{\beta}_1 = \frac{4x_{11}y_1 - 4x_{11}^2\hat{\beta}_2}{4x_{11}^2 + 2\lambda}$$

$$\frac{\partial f}{\partial \hat{\beta}_2} = -4x_{11}y_1 + 4x_{11}^2\hat{\beta}_2 + 4x_{11}^2\hat{\beta}_1 + 2\lambda\hat{\beta}_2 = 0$$

$$4x_{11}^2\hat{\beta}_2 + 2\lambda\hat{\beta}_2 = 4x_{11}y_1 - 4x_{11}^2\hat{\beta}_1$$

$$\hat{\beta}_2 = \frac{4x_{11}y_1 - 4x_{11}^2\hat{\beta}_1}{4x_{11}^2 + 2\lambda}$$

let $A = \frac{4x_{11}y_1}{4x_{11}^2+2\lambda}$ and $B = -\frac{4x_{11}^2}{4x_{11}^2+2\lambda}$

then $\hat{\beta}_1 = A + B\hat{\beta}_2$ and $\hat{\beta}_2 = A + B\hat{\beta}_1$

Substituting, we arrive at $\hat{\beta}_1 = A + B(A + B\hat{\beta}_1)$

then $\hat{\beta}_1 = A + AB + B^2\hat{\beta}_1$

$$\hat{\beta}_1 - B^2\hat{\beta}_1 = A + AB$$

$$\hat{\beta}_1 = \frac{A + AB}{1 - B^2}$$

furthermore, $\hat{\beta}_2 = A + B(A + \hat{\beta}_2)$, and similarly

$$\hat{\beta}_2 = \frac{A + AB}{1 - B^2}$$

therefore $\hat{\beta}_1 = \hat{\beta}_2$. ☐

c. the lasso optimization problem is to minimize

$$\sum_{i=1}^{2}\left(y_i - \beta_0 - \sum_{j=1}^{2}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{2}|\beta_j|$$

similarly to parts a and b, in this setting

the equation would be

$$2(y_1 - \hat{\beta}_1 x_{11} - \hat{\beta}_2 x_{11})^2 + \lambda(|\hat{\beta}_1| + |\hat{\beta}_2|).$$

alternatively, we want to minimize

$$2(y_1 - (\hat{\beta}_1 + \hat{\beta}_2) x_{11})^2 \geq 0 \text{ with } |\hat{\beta}_1| + |\hat{\beta}_2| \leq s.$$

d. to minimize $2(y_1 - (\hat{\beta}_1 + \hat{\beta}_2) x_{11})^2$, $\hat{\beta}_1 + \hat{\beta}_2 = \frac{y_1}{x_{11}}$, which results in $2(y_1 - (\hat{\beta}_1 + \hat{\beta}_2) x_{11})^2 = 0$. solutions $(\hat{\beta}_1, \hat{\beta}_2)$ occur along the line $\hat{\beta}_2 = -\hat{\beta}_1 + \frac{y_1}{x_{11}}$, and a minimal solution occurs where $\hat{\beta}_2 = -\hat{\beta}_1 + \frac{y_1}{x_{11}}$ touches the constraint region $|\hat{\beta}_1| + |\hat{\beta}_2| \leq s$. since $|\hat{\beta}_1| + |\hat{\beta}_2| = s$ is a line with equation $\hat{\beta}_2 = -\hat{\beta}_1 + s$ in quadrant I, the constraint line and solution line are parallel. therefore there are an infinite number of points where $(\hat{\beta}_1, \hat{\beta}_2)$ satisfy the optimization problem within the constraint. ∎

8.4.5

majority vote approach:

if $P < 0.5$, that bootstrap sample predicts green.

if $P \geq 0.5$, that bootstrap sample predicts red.

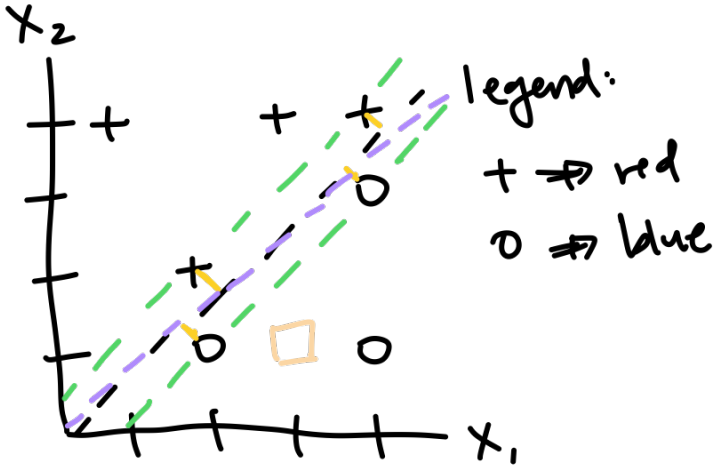
therefore we have green × 4, red × 6 ⇒ final classification red.

average probability approach:

the average probability is $0.45 < 0.5$ ⇒ final classification green.

9.7.3

a.



b. sketched with a dashed line above.

since observations 2 and 3 are on the line $X_2 = X_1$ and observations 5 and 6 are on the line $X_2 = X_1 - 1$, the optimal separator will be $X_2 = X_1 - 0.5$.

c. the graph shows that observations are red when $X_2 > X_1 - 0.5$ and blue elsewise. then the classification rule for this separator is classify red if $X_2 - X_1 + 0.5 > 0$, and classify blue otherwise. therefore, $B_0 = 0.5$, $B_1 = -1$, $B_2 = 1$.

d. the margin is indicated in green.

e. the support vectors are indicated in yellow.

f. since observation 7 is not a support vector, we know from the textbook that a slight movement from observation 7

will not change the margin. so long as that movement does not bring it within the current margin; but since the margin is not dependent on observation 7, its movement does not necessarily mean the margin will change.

g. indicated in purple. the equation for this hyperplane is $X_2 = \frac{4}{5} X_1$.

h. indicated as a light orange square.